

Quantum computing for the quantum-curious

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Phasecraft

What I will discuss today

This is meant to be a non-technical talk to familiarize you with the quantum computer sector, its motivations, current priorities, and future directions.

- Why are we building quantum computers and designing quantum algorithms?
- Who is building them?
- What is the current state of the technology?
- Roles of the key actors within the sector
- Where are things headed in the next decade?

Content is intentionally explained at a high(ish) level and doing some accuracy<>understandability tradeoffs.

Throughout, I will point to helpful resources that you can consult if interested in learning more about a particular topic.

What I will not discuss today (but my colleague Marcos Marcos will)

Quantum mechanics

Computational complexity

How quantum mechanics is exploited for useful computations



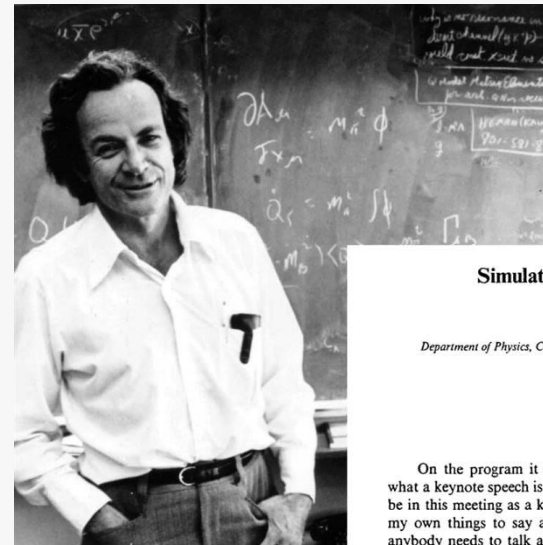
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Why do we want to build quantum computers?

Feynman's 1982 argument: **nature is quantum-mechanical, so a classical computer cannot simulate it efficiently.**

The cost of tracking a quantum system on a classical machine grows exponentially with the number of particles.

- A computer built from quantum parts could simulate quantum systems directly, sidestepping that exponential cost.
- The target: chemistry and materials, where the behavior of electrons in a molecule or solid is governed by the same quantum rules the machine would run on.



Simulating Physics with Computers

Richard P. Feynman

Department of Physics, California Institute of Technology, Pasadena, California 91107

Received May 7, 1981

1. INTRODUCTION

On the program it says this is a keynote speech—and I don't know what a keynote speech is. I do not intend in any way to suggest what should be in this meeting as a keynote of the subjects or anything like that. I have my own things to say and to talk about and there's no implication that anybody needs to talk about the same thing or anything like it. So what I want to talk about is what Mike Dertouzos suggested that nobody would talk about. I want to talk about the problem of simulating physics with computers and I mean that in a specific way which I am going to explain. The reason for doing this is something that I learned about from Ed Fredkin, and my entire interest in the subject has been inspired by him. It has to do with learning something about the possibilities of computers, and also something about possibilities in physics. If we suppose that we know all the physical laws perfectly, of course we don't have to pay any attention to computers. It's interesting anyway to entertain oneself with the idea that we've got something to learn about physical laws; and if I take a relaxed view here (after all I'm here and not at home) I'll admit that we don't understand everything.

The first question is, What kind of computer are we going to use to simulate physics? Computer theory has been developed to a point where it realizes that it doesn't make any difference; when you get to a *universal computer*, it doesn't matter how it's manufactured, how it's actually made. Therefore my question is, Can physics be simulated by a universal computer? I would like to have the elements of this computer *locally interconnected*, and therefore sort of think about cellular automata as an example (but I don't want to force it). But I do want something involved with the

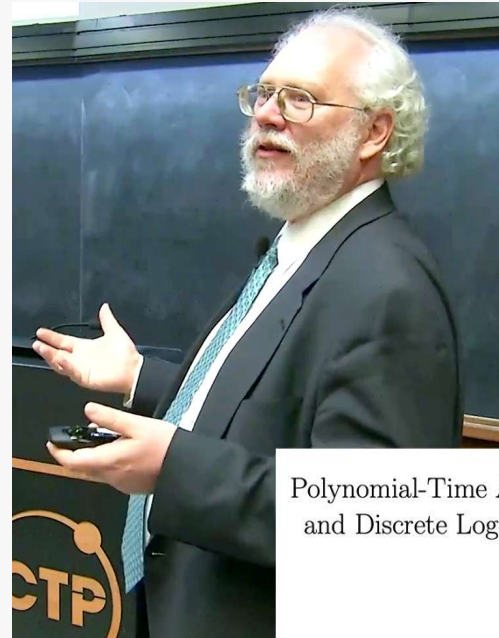
Simulation Physics with Computers:

<https://s2.smu.edu/~mitch/class/5395/papers/feynman-quantum-1981.pdf>

Why do we want to build quantum computers?

In 1994 Peter Shor showed a **quantum computer could factor large numbers and compute discrete logarithms in hours rather than the billions of years** a classical machine would take.

- The algorithm breaks RSA and elliptic-curve cryptography, which protect most internet traffic, banking, and stored data.
- It turned quantum computing from a physics-simulation idea into a national-security concern and initiated sustained government funding.
- Shor's algorithm needed a machine that did not exist (and still doesn't).



Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer*

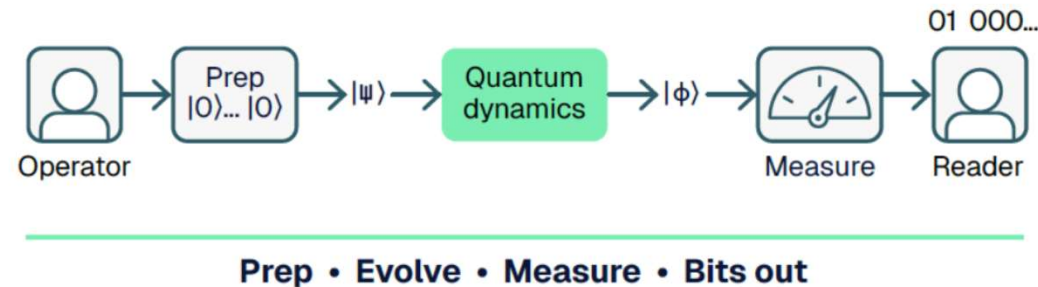
Peter W. Shor[†]

Abstract

A digital computer is generally believed to be an efficient universal computing device; that is, it is believed able to simulate any physical computing device with an increase in computation time by at most a polynomial factor. This may not be true when quantum mechanics is taken into consideration. This paper considers factoring integers and finding discrete logarithms, two problems which are generally thought to be hard on a classical computer and which have been used as the basis of several proposed cryptosystems. Efficient randomized algorithms are given for these two problems on a hypothetical quantum computer. These algorithms take a number of steps polynomial in the input size, e.g., the number of digits of the integer to be factored.

Quantum Computers

(at a high level)



A quantum computer does three things in order:

- It **prepares** a carefully chosen starting state on a set of quantum bits (qubits).
- It lets that state **evolve** under precise physical interactions: the **quantum dynamics**.
- Then it **measures** the result, turning quantum information back into ordinary bits: a string of 0s and 1s you can read on a screen.

Because it **uses quantum dynamics rather than classical operations**, it can change states in many ways (including processing many possibilities at once) that classical computers cannot easily reproduce.

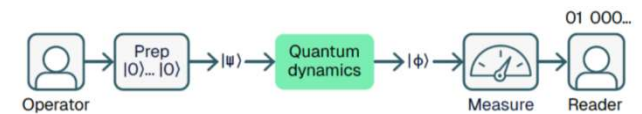
Quantum Computers

(at a high level)

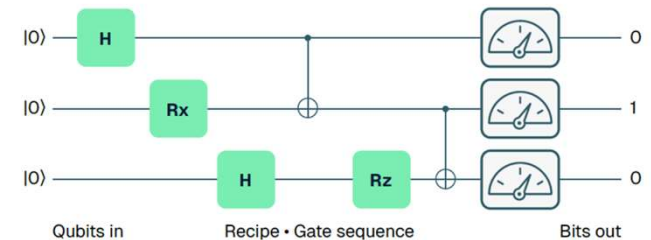
A **quantum algorithm** controls the middle step via well-defined **sequence of gate operations**.

Rotations and entangling interactions are applied to the qubits in a specific order. You then repeat the run multiple times to get an output with the required confidence level.

Designing a good quantum algorithm means finding methods to useful answers faster, more accurately, or at a lower cost than any classical method we know.



Prep • Evolve • Measure • Bits out



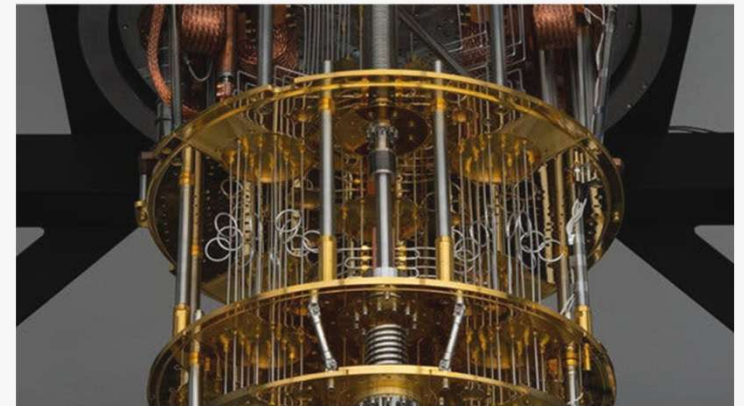
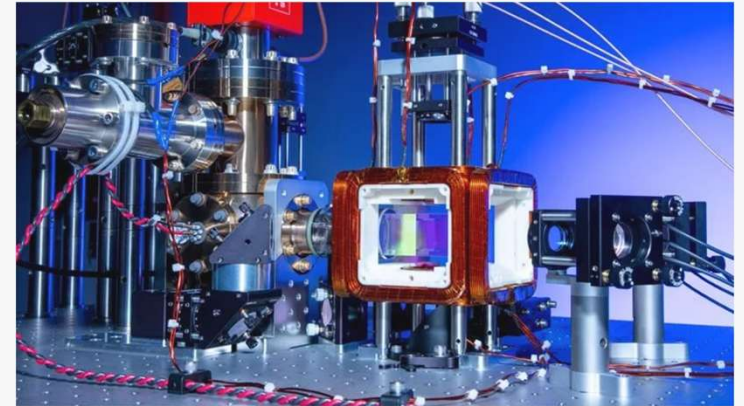
The recipe that drives the dynamics box above

The current state of quantum hardware

We are currently in the **NISQ era: Noisy Intermediate-Scale Quantum**.

Machines have hundreds to ~1,000+ physical qubits, but every operation carries error, which limits how long a computation can run before noise swamps it.

Overhead is steep: with most error correcting codes you need hundreds of **physical qubits** per **reliable logical qubit**.



Who is building quantum hardware?

Many efforts* are underway! The picture is slightly complicated by the fact that quantum computers come in many flavors, with each having very distinct characteristics, capabilities, and trade-offs.

Superconducting

Fast gates and mature fabrication. Qubits decohere quickly (~milliseconds) and need cooling near absolute zero.



Trapped Ions

Highest gate fidelities and long coherence, with all-to-all connectivity. Slower gate speeds limit throughput.



Neutral Atom

Large arrays (1,000+ atoms) and flexible, reconfigurable connectivity. Gate speeds and readout still maturing.



Silicon Spin

Built on semiconductor fabrication, so potentially scalable. Less mature so fidelities and qubit counts lag other platforms.



Photonic

Room-temperature operation (for the most part), good for networking. Photons are easily lost, which drives up the error-correction burden



*this is not an exhaustive list.. several more companies active in developing each architecture type, and some newer modalities like nitrogen-vacancy diamond have also been proposed

What about algorithms and software*?

Many (or most) hardware companies have small in-house algorithms teams. But dedicated efforts in algorithms are mostly in academia, national labs, and a few early-stage startups.



P H A S E C R A F T

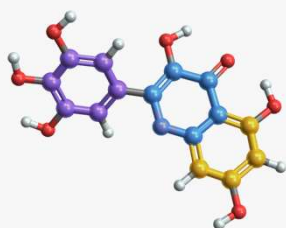
my employer



*there is a whole middle layer of quantum software for things like hardware control, compilation, error correction, etc. that will not be discussed since its mostly irrelevant for the purpose of this talk

The current state of algorithms

(and where we think quantum advantages may exist)



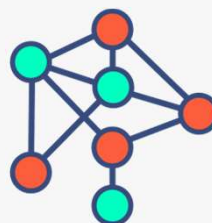
Simulating molecules and materials

Direct simulation of quantum systems: the original rationale with advantages well grounded in theory.



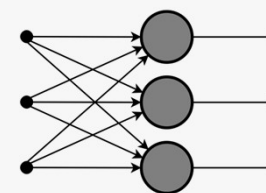
Cryptography

Shor's algorithm gives an exponential speedup for factoring and discrete logs. Provable and the motivation for transitioning to post-quantum cryptography.



Optimization

Some speedups exist (e.g. QAOA), but setup costs often erase them in practice. Advantage is case-by-case, not settled. Hybrid methods* may be the path to near-term advantage



Machine Learning

Speedups appear to be modest, unproven at scale, and may be matched by classical methods. Some recent papers show great promise.** An open research question.

* Quantum-Enhanced Optimization by Warm Starts: <https://arxiv.org/html/2508.16309v1>

** Exponential quantum advantage in processing massive classical data: <https://arxiv.org/pdf/2604.07639>

The government's role within the Quantum Ecosystem

DARPA's Quantum Benchmarking Initiative

Flagship government program on quantum computing seeking to answer:
“will a utility scale quantum computer exist by 2033”

- It is not a competition: DARPA is evaluating most viable approaches being pursued by industry, using an independent test-and-evaluation team
- Three stage process for hardware companies:
 - Stage A: describe a full system concept and show feasibility
 - Stage B: deliver a detailed R&D roadmap to achieving utility-scale capabilities
 - Stage C: independent verification of hardware
- In November 2025, 11 companies advanced to Stage B across superconducting, trapped-ion, neutral-atom, and photonic platforms.



The government's role within the Quantum Ecosystem



ARPA-E Quantum Computing for Computational Chemistry

Quantum algorithms for chemistry and materials in the energy sector. Every recipient must run its solution on real quantum hardware. US government's first applications-driven quantum program.



National Science Foundation

Foundational quantum information science research at multiple scales (small awards to large center-scale projects).



U.S. DEPARTMENT
of **ENERGY**

Department of Energy

DOE runs five National QIS Research Centers led by national labs (Argonne, Brookhaven, Fermilab, Oak Ridge, Lawrence Berkeley).



More recently... the Department of Commerce announced its intent to provide funding to 9 companies in exchange for equity for the development of quantum hardware and chip fabrication

There are also other government programs supporting QIS work, so not a comprehensive list

Both hardware and algorithms are maturing fast

Hardware is advancing towards fault tolerance:

- Companies like Quantinuum are now demonstrating error-corrected (logical) qubits.
- Google's Willow (105 physical qubits) showed **below-threshold error correction**, where adding more physical qubits to protect one logical qubit lowered its error rate rather than raising it. Necessary signature for scaling to larger systems.
- Some companies like PsiQuantum and Alice & Bob are entirely bypassing NISQ hardware and aiming straight for fault tolerance

Both hardware and algorithms are maturing fast

Algorithms are becoming increasingly more efficient

- Phasecraft* demonstrated that you can cut the circuit depth for simulating materials so it no longer grows with system size, e.g. reducing the gate count for simulating a lithium-ion battery material from ~1.5 trillion to ~410,000.
- On cryptography, Google's** 2025 estimate lowered the cost of breaking RSA-2048 from 20 million physical qubits (2019) to under 1 million.
- In 2026*** a Google, Ethereum Foundation, and Stanford team put breaking Bitcoin's elliptic-curve signatures at under 1,200 logical qubits.

Material	Structure / role	Applications	Baseline qubits	Magritte qubits	Baseline circuit depth	Magritte Circuit depth
SrVO₃	Cubic perovskite (correlated oxide)	Batteries, fuel cells	864	180	6.69×10^8	880
Li₂CuO₂	Battery cathode	Batteries	1,260	1,120	2.09×10^9	7,900
GaAs	Zincblende semiconductor	Transistors, photovoltaics	1,500	1,024	3.51×10^9	8,394
Si	Diamond-cubic semiconductor	Batteries, catalysis, LEDs, microelectronics, PV	750	1,120	4.26×10^8	8,507
SrZrO₃	Tetragonal perovskite	Batteries, catalysis, fuel cells, LEDs	864	1,098	7.54×10^8	11,431

* <https://www.nature.com/articles/s41467-023-43479-6>

** <https://arxiv.org/abs/2505.15917>

*** <https://arxiv.org/abs/2603.28846>

What are the industry's main goals for the next decade?

Hardware: can we achieve fault tolerance? scale into large enough systems that provide utility?

- **Reach fault tolerance:** error rates low enough that logical qubits run arbitrarily long computations. IBM targeting ~200 logical qubits by 2029; several vendors target thousands into the early 2030s.
- Scale the **physical-to-logical overhead** down and hold gate fidelity as systems grow.
- Cut the **cost per logical operation** so a machine's value can exceed its cost

What are the industry's main goals for the next decade?

Applications: can utility be achieved in the NISQ era? what can we do with a fault-tolerant system that is scientifically and commercially useful?

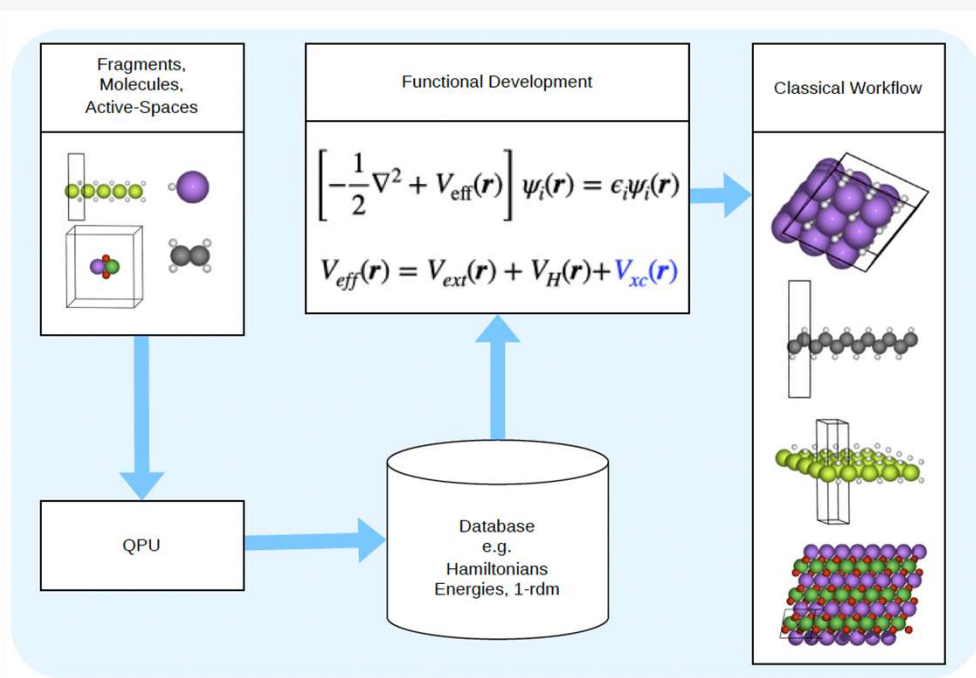
- NISQ era: find problems where today's noisy hardware beats the best classical method. We are already starting to see the first demonstrations of **scientific quantum advantage**. Commercial or “practical” quantum advantage not yet evident.
- Fault-tolerant era: well defined applications in **cryptography and simulating quantum systems** (chemistry, materials).
- **Optimization and machine learning** are open areas for identifying quantum advantage

Some applications under pursuit at Phasecraft

Materials and Molecular Modeling: end-to-end compiler that takes an atomic-level specification of a target system and produces an optimized quantum circuit for calculating ground-state energies and time-dynamics simulations

Materials and Molecular Simulation: delegating the classically hard electronic structure calculations within computational chemistry workflows to the quantum computer

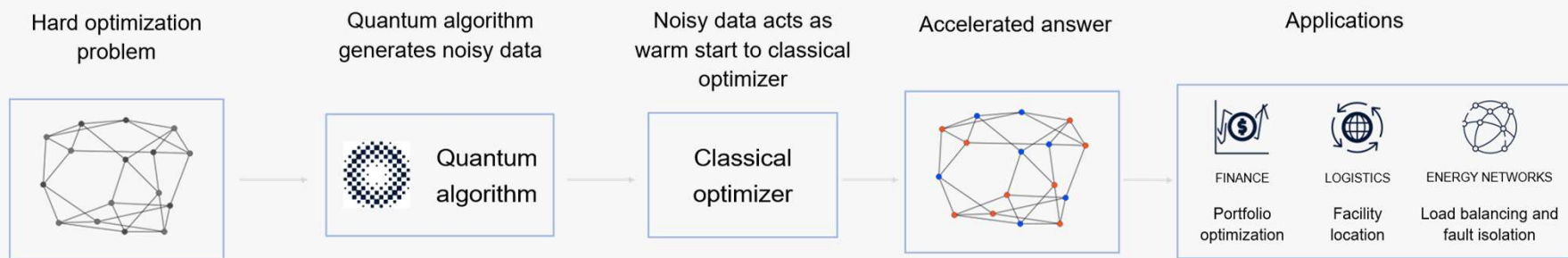
Workflow of Quantum-Enhanced Density Functional Theory (QEDFT)



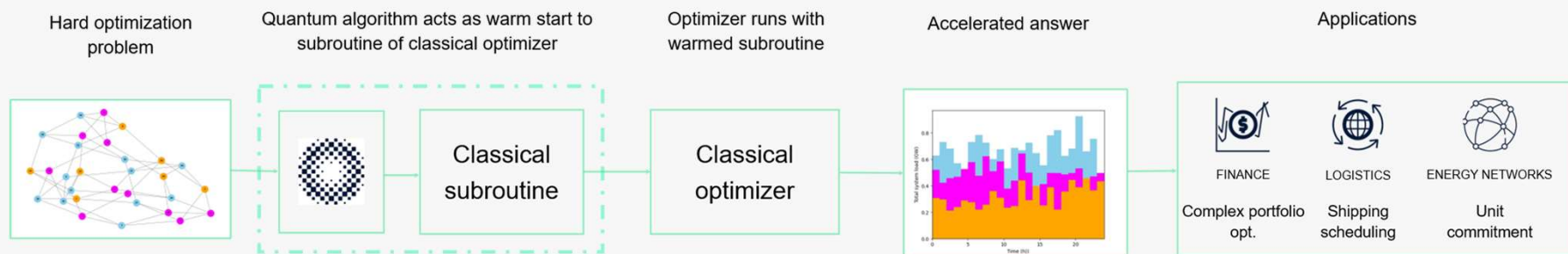
Some applications under pursuit at Phasecraft

Quantum-Enhanced Optimization: using quantum computation as a warm-start for speeding up classical solvers; will have more to share soon on applications of this to branch and bound problems

For optimization problems formulated like Max-Cut



For many other optimization problems (e.g. Mixed Integer Linear Programming)



Quantum Computing for Hydrogen Fuel Production



PHASECRAFT

Washington, DC; UK: London, Bristol

Project Lead; Quantum algorithms and software



Johnson Matthey
Inspiring science, enhancing life

Wayne, PA; UK: Sonning

Catalyst expertise; classical simulation and industrial validator



Boston, MA

Neutral-atom hardware implementation

HARVARD
UNIVERSITY

Cambridge, MA

Pathfinding hardware experiments

Quantum-enhanced simulation methods to model metal oxide and chalcogenide catalysts for electrolytic hydrogen production, targeting quantum integration within industrial materials screening workflows

SELECTED HARDWARE
PROVIDER

QuEra Computing (neutral atoms); IBM (superconducting)

ENABLING FEATURES

co-designed algorithms that cut quantum resource requirements for strongly correlated materials, hardware-aware compilation for neutral atoms, and integration with existing classical tools used in the energy sector

GOAL

scalable quantum algorithms for materials modeling and simulation targeting catalysts, implemented on hardware and validated against classical and experimental benchmarks

The grand challenge of quantum applications

The Google Quantum AI team recently published a very thoughtful perspective on quantum applications and a framework for the sector. Some of their key observations and recommendations:

- The paper proposes a **five-stage framework** from an abstract algorithm to a deployed, advantage-bearing application.
- Progress is uneven across the stages. The field is strong at Stage I (discovering abstract algorithms) and Stage IV (resource estimation for a few select use cases), but underinvested and weak in the middle:
 - **Stage II:** finding concrete hard instances where quantum speedups survive real world constraints
 - **Stage III:** connecting to real applications
- **Cryptanalysis and quantum simulation** are the only two domains where the Stage III connection has been made: everywhere else, an abstract speedup exists but link to real-world applications is not as clearly defined.
- A useful quantum application should also **resist classical mimicry**: if a classical model can learn to reproduce its outputs, the advantage erodes.
- The call to the field: match hardware progress with equal investment in **algorithms and in identifying the problems worth pointing them at**.

Link to the Google paper:

<https://arxiv.org/abs/2511.09124>

NSF Workshop on Quantum Algorithms for CPS



Focus on quantum algorithms for optimization with applications in transportation networks, energy grids, advanced manufacturing, and autonomous systems

Planned for the week of January 13, 2027, at Phasecraft's US Office in the DC area

NSF Award 2628842

https://www.nsf.gov/awardsearch/show-award?AWD_ID=2628842

Other helpful literature to learn more

Thinking Inside the Box by Alice & Bob: <https://alice-bob.com/wp-content/uploads/2024/12/Think-Inside-The-Box-Alice-Bob-Whitepaper.pdf>

Parts 1 and 2 are great for learning about quantum computing.

The [**Quantum Economic Development Consortium**](#) also has lots of helpful content on its website, including a [report on the intersection of Quantum and AI compute](#) and [Quantum Computing for Transportation and Logistics](#)

The US Department of Transportation published a report (following a workshop) in 2024 on [**Quantum Technologies for the Transportation Sector**](#)

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